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Studierichting: Informatica
Oefeningenreeks 1B nr. 37.6

$$5 - 12i = (x + yi)^2 = x^2 - y^2 + 2xyi$$

$$\begin{cases} x^2 - y^2 = 5 \\ 2xy = -12 \end{cases} \Rightarrow \begin{cases} x^2 - y^2 = 5 \\ xy = -6 \end{cases} \Rightarrow \begin{cases} x^2 - y^2 = 5 \\ x^2(-y)^2 = 36 \end{cases}$$

$$\lambda^2 - 5\lambda - 36 = 0$$

$$\Delta = (-5)^2 - 4 \cdot (-36) = 25 + 144 = 169 = 13^2$$

$$\lambda_1 = \frac{5 + 13}{2} = 9$$

$$\lambda_2 = \frac{5 - 13}{2} = -4$$

$$x = \pm 3$$

$$y = \pm 2$$

$$x - yi \in \{3 - 2i, -3 + 2i\}$$

References

- [1] code template gebruikt van Tiffany Kegels 1B-37.5